

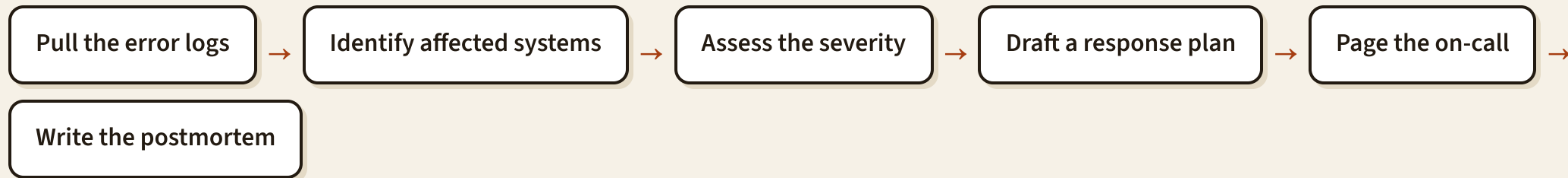
Architecture Over Intelligence

Designing reliable AI workflows

Amar Patel · The AI Fellowship Madrid · 9 Oct 2026



"Investigate this incident" is one job. *It isn't.*



One request sets off a chain of small jobs.

Each job needs the one before it. If one step drifts, every step after it drifts too.

The demo vs. production gap

Demo

- One uninterrupted session
- Mistakes are annoying
- A human watches the whole time
- A frontier model

Production

- Compaction, reset, walk-aways
- Mistakes are data loss
- A human checks hours later
- A cheaper model that drifts

A demo works because nothing interrupts it.

In production, the session ends, the model changes, and a mistake now costs you.

A JSON file is the *source of truth*

● process-state.json		meeting_id: 2026-01-15_140000
current_step	2.6 → next is 3	
completed_steps	[-1, 0, 1, 2, 2.5, 2.6]	
user_confirmed.summary	true	
aborted	false	

Every step reads the same file before it runs, and the file says exactly where you are.

If the session dies, the next run starts from that same file and that same step.

A gate is *authority*, not just quality control



Auto

Safe, predictable steps keep moving.
File reads, health checks.



Human

The agent stops and waits.
Anything that needs a yes.



Blocked

Cannot proceed.
Missing file, corrupted state.

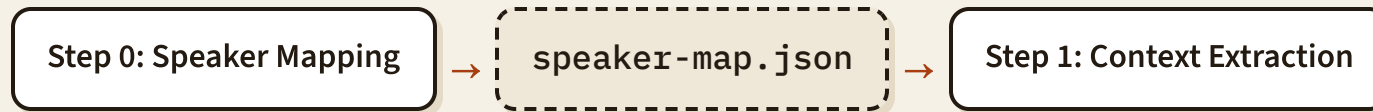


REVIEWS EVERY STEP

GUARDS ONLY THE IRREVERSIBLE

You don't remove the human as the model improves. You move it: from checking every step, to guarding only the steps that cannot be undone.

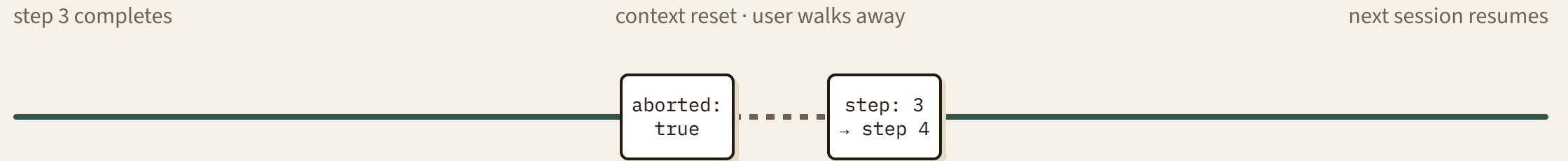
No step uses data that doesn't exist yet



Each step says what it creates, and the next step checks that file really exists before it runs.

A step that is missing its input stops and tells you, instead of guessing.

Interruption is not a bug



A single file remembers where the work stopped.

It survives a crash, a context reset, and you walking away.

One real pipeline: *11 steps, 8 LLM calls*



9

HUMAN CHECKPOINTS

4

SCRIPTS

1

STATE FILE

Nine of the 11 steps stop for a human; two do not.

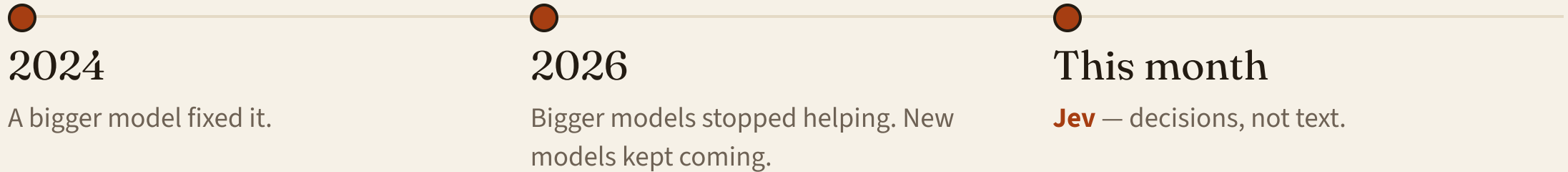
What broke in three months

- ✎ The state file got edited to the wrong step → *validate on load*
- ⏮ The LLM skipped a gate → *the file decides, not the prompt*
- ✓ Compaction mid-pipeline → *the file survived* — the win

Trust the file, and check the artifact itself.

Never trust what the previous step said it did.

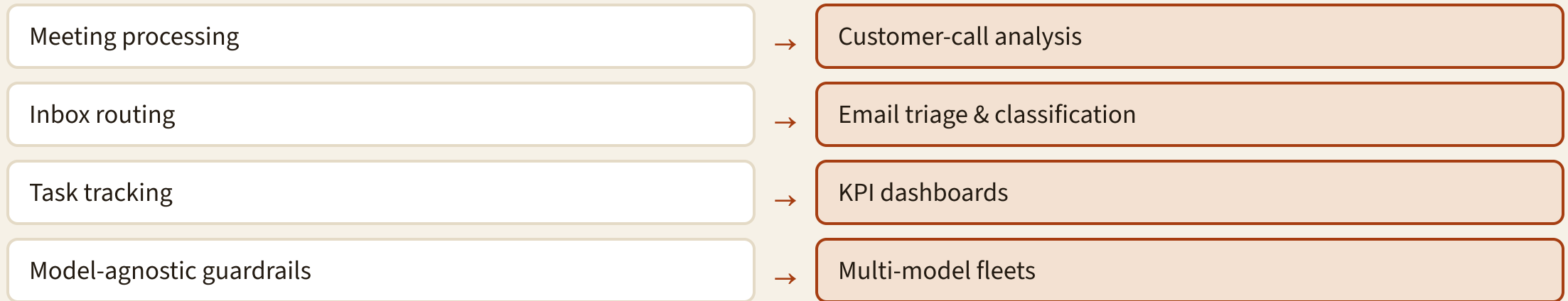
The model is always changing



“The State Machine Is the Agent.”

The model changes every month. What stays reliable is the structure around it, not the model itself.

These patterns work everywhere



The kind of work changes. The pattern stays the same.

One pattern, many jobs: customer calls, email triage, KPI reports.

Five rules

- 1 The **state file** is the source of truth.
- 2 **Gates** are structural, not optional.
- 3 Declare **artifacts** before they're needed.
- 4 Design for **interruption**.
- 5 **Architecture > intelligence.**

Key takeaways

- The model changes. The structure stays.
- Better models do not remove the human; they move it to the irreversible steps.
- The same structure works with any model. That part comes free.



REVIEWS EVERY STEP

GUARDS ONLY THE IRREVERSIBLE

Reliability at the boundaries.